Transfer Learning Through Conditional Quantile Matching

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Central Applied Science

Meta

Business Question: How do Meta's family of apps perform compared to an app developed by a competing company?



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Objective: Predict the country-level MAU of a target app using data from Meta's family of apps.

Regression Task: Predict the country-level MAU $Y^{(0)} \in \mathcal{Y}$ for a target app from country-specific covariates $X^{(0)} \in \mathcal{X}$.

1. **Source-Domain Data:** $\mathcal{D}_S^{(k)} = \left\{ \left(X_i^{(k)}, Y_i^{(k)} \right) \right\}_{i=1}^{n_k} \sim P^{(k)}$ for $k = 1, \ldots, K$.
2. **Target-Domain Data:** $\mathcal{D}_T = \left\{ \left(X_i^{(0)}, Y_i^{(0)} \right) \right\}_{i=1}^{n_0} \sim P^{(0)}$, where $n_0 \ll n_k$ for $k = 1, \ldots, K$.

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Covariate Shift (Shimodaira, 2000):
$P^{(k)}(Y|X) = P^{(0)}(Y|X)$ but $P^{(k)}(X) \neq P^{(0)}(X)$.

Label Shift (Saerens et al., 2002):
$P^{(k)}(X|Y) = P^{(0)}(X|Y)$ but $P^{(k)}(Y) \neq P^{(0)}(Y)$.

General Target Shift:

$$P^{(k)}(X) \neq P^{(0)}(X) \quad \text{and} \quad P^{(k)}(Y|X) \neq P^{(0)}(Y|X) \quad \text{for } k = 1, \ldots, K.$$

Proposed Transfer Learning Framework (TLCQM)

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1. Learn a generative model $\widehat{P}^{(k)}(\cdot|x)$ using $\left\{\left(X_i^{(k)}, Y_i^{(k)}\right)\right\}_{i=1}^{n_k}$ for each source domain k.
   - Model $P^{(k)}(\cdot|x) = g^{(k)}(x, \cdot)_{\#}P_\eta$ and estimate $\widehat{g}^{(k)}$ via engression (Shen and Meinshausen, 2025):

$$\widehat{g}^{(k)} \in \underset{g \in \mathcal{G}}{\arg\min} \frac{1}{n_k} \sum_{i=1}^{n_k} \left[\frac{1}{m} \sum_{j=1}^{m} \left| Y_i^{(k)} - g(X_i^{(k)}, \eta_{ij}) \right| - \frac{1}{2m(m-1)} \sum_{j=1}^{m} \sum_{j'=1}^{m} \left| g(X_i^{(k)}, \eta_{ij}) - g(X_i^{(k)}, \eta_{ij'}) \right| \right],$$

   where P_η is a prespecified noise distribution and $\mathcal{G}$ is a neural network class.

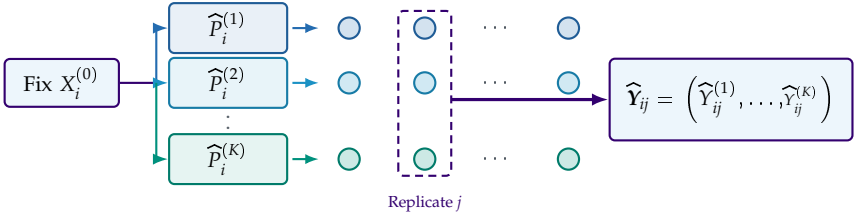
1 Learn source distributions
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$\longrightarrow$

2 Generate at target covariates
$\widehat{Y}_{ij}^{(k)} \sim \widehat{P}^{(k)}(\cdot|X_i^{(0)})$ for $j = 1, \ldots, M$.

$\longrightarrow$

2. For each target covariate $X_i^{(0)}$, independently draw M synthetic response vectors:



$$\left\{\left(X_i^{(0)}, \widehat{\mathbf{Y}}_{ij}\right) : i = 1, \ldots, n_0,\ j = 1, \ldots, M\right\}, \quad \widehat{\mathbf{Y}}_{ij} = \left(\widehat{Y}_{ij}^{(1)}, \ldots, \widehat{Y}_{ij}^{(K)}\right), \quad \widehat{Y}_{ij}^{(k)} \sim \widehat{P}^{(k)}\left(\cdot|X_i^{(0)}\right).$$

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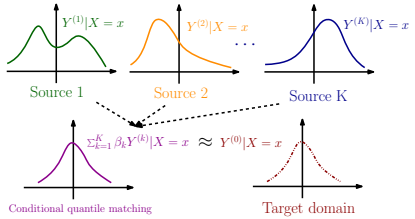
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3 Conditional quantile calibration
A shared linear combination minimizes empirical CRPS.

3. Compute the conditional quantile calibration estimator by minimizing the continuous ranked probability score (CRPS):

$$\widehat{\boldsymbol{\beta}} \in \arg\min_{\boldsymbol{\beta}\in\Theta} \frac{1}{n_0}\sum_{i=1}^{n_0}\left\{\frac{1}{M}\sum_{j=1}^{M}\left|Y_i^{(0)} - \boldsymbol{\beta}^T\widehat{\boldsymbol{V}}_{ij}\right| - \frac{1}{2M(M-1)}\sum_{j,j'=1}^{M}\left|\boldsymbol{\beta}^T\widehat{\boldsymbol{V}}_{ij} - \boldsymbol{\beta}^T\widehat{\boldsymbol{V}}_{ij'}\right|\right\}.$$



- $\widehat{\boldsymbol{V}}_{ij} = (1, \widehat{\boldsymbol{Y}}_{ij})^T$, and $Y_i^{(0)}$ is matched with draws generated at the same target covariate $X_i^{(0)}$.
- $\Theta = \left\{\boldsymbol{\beta} \in \mathbb{R}^{K+1} : ||\boldsymbol{\beta}||_1 \leq B_\beta\right\}$.

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4 Augment and fit on $\mathcal{D}_T \cup \mathcal{D}_A$
XGBoost, kernel ridge regression (KRR), neural network (NN),...

4. Augment the target domain with synthetic, target-calibrated labels:

$$\mathcal{D}_A = \bigcup_{k=1}^{K} \left\{ \left(X_i^{(k)}, \widehat{\boldsymbol{\beta}}^T \widehat{\boldsymbol{V}}_i^{(k)} \right) \right\}_{i=1}^{n_k} \quad \text{with} \quad \widehat{\boldsymbol{V}}_i^{(k)} = \left(1, \widehat{Y}_i^{(1,k)}, \ldots, \widehat{Y}_i^{(K,k)}\right)^T \in \mathbb{R}^{K+1},$$

where $\widehat{Y}_i^{(j,k)}$ is a prediction of $Y^{(j)}$ at $X_i^{(k)}$, such as $\widehat{\mathbb{E}}(Y^{(j)}|X_i^{(k)})$.

5. (Optional) Estimate $w_k(x) = \frac{dP_X^{(0)}(x)}{dP_X^{(k)}(x)}$ for $k = 1, \ldots, K$ to correct for covariate shift.

Theory: Excess Risk Decomposition

- $R(f) = \mathbb{E}_{P^{(0)}}[\ell(Y^{(0)}, f(X^{(0)}))]$ and $f^{(0)} \in \arg\min_{f \in \mathcal{F}} R(f)$ under a loss function ℓ.
- $\mathrm{Rad}_n(\mathcal{F})$ is the Rademacher complexity of $\mathcal{F}$.
- Define the conditional CRPS as $\widehat{\Delta}(\widehat{\boldsymbol{\beta}}) := \mathbb{E}_{X^{(0)}}\left\{\int_{\mathbb{R}} \left[F^{(0)}(t|X^{(0)}) - \widehat{G}_{\widehat{\boldsymbol{\beta}}}(t|X^{(0)})\right]^2 dt\right\}$.

Target-Only: $\widehat{f}^{(0)} = \arg\min_{f \in \mathcal{F}} \frac{1}{n_0} \sum_{i=1}^{n_0} \ell\left(Y_i^{(0)}, f(X_i^{(0)})\right)$ satisfies

$$R(\widehat{f}^{(0)}) - R(f^{(0)}) \lesssim \mathrm{Rad}_{n_0}(\mathcal{F}) + \sqrt{\frac{\log(2/\delta)}{n_0}} \quad \text{w. p. } \geq 1 - \delta.$$

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TLCQM: $\widehat{f}^{(0,tl)} = \arg\min_{f \in \mathcal{F}} \frac{1}{N} \left\{\sum_{i=1}^{n_0} \ell\left(Y_i^{(0)}, f(X_i^{(0)})\right) + \sum_{k=1}^{K} \sum_{i=1}^{n_k} \widehat{w}_k(X_i^{(k)}) \cdot \ell\left(\widehat{\boldsymbol{\beta}}^T \widehat{\boldsymbol{V}}_i^{(k)}, f(X_i^{(k)})\right)\right\}$ satisfies

$$R(\widehat{f}^{(0,tl)}) - R(f^{(0)}) \lesssim \underbrace{\mathrm{Rad}_N(\mathcal{F}) + \sqrt{\frac{\log(1/\delta)}{N}}}_{\text{Standard generalization error}} + \underbrace{\frac{1}{N}\sum_{k=1}^{K} \|\widehat{w}_k - w_k\|_1}_{\text{Importance weight error}} + \underbrace{\frac{(N - n_0)^2 \cdot \widehat{\Delta}(\widehat{\boldsymbol{\beta}})}{N^2}}_{\text{Conditional transfer error}}$$

w. p. $\geq 1 - \delta$, where $N = \sum_{k=0}^{K} n_k$.

Theory: Conditional Quantile Calibration Error

Let $\boldsymbol{\beta}_* \in \arg\min_{\boldsymbol{\beta}\in\Theta} \Delta(\boldsymbol{\beta})$, where $\Delta(\boldsymbol{\beta})$ is the population conditional CRPS discrepancy.

$$\widehat{\Delta}(\widehat{\boldsymbol{\beta}}) \lesssim \underbrace{\Delta(\boldsymbol{\beta}_*)}_{\text{Transfer bias}} + \underbrace{\sum_{k=1}^{K} \|\widehat{g}^{(k)} - g^{(k)}\|_\infty}_{\text{Source generator error}} + \underbrace{\sqrt{\frac{[\log n_0 + \log(1/\delta)]\left[\Delta(\boldsymbol{\beta}_*) + \sum_{k=1}^{K} \|\widehat{g}^{(k)} - g^{(k)}\|_\infty\right]}{n_0}}}_{\text{Localized target-sample error}}$$

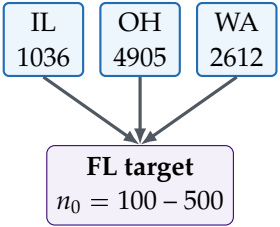
$$+ \underbrace{\frac{\log n_0 + \log(1/\delta)}{n_0} + \epsilon_{\text{opt}} + O\left(\frac{1}{\sqrt{M}}\right)}_{\text{Optimization and Monte Carlo approximation errors}} .$$

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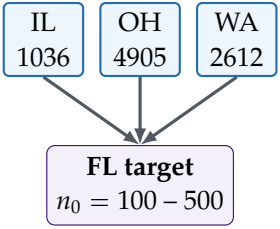
1. $\Delta(\boldsymbol{\beta}_*)$ is the **irreducible transfer bias**: the best conditional distribution approximation attained by the calibrated source family.
2. It vanishes when $P^{(0)}(Y|X)$ lies in the **affine span** of the source conditional laws.
3. Large $\Delta(\boldsymbol{\beta}_*)$ can outweigh variance reduction and cause **negative transfer**.

Goal: Predict apartment rental prices in FL using data from IL, OH, and WA.



500 Monte Carlo replications.

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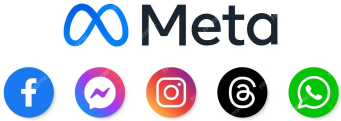
500 Monte Carlo replications.

MSE reduction versus target-only training

n_0	100	200	300	500
XGBoost	12.0%	5.5%	1.7%	−2.7%
KRR	**78.5%**	**66.0%**	**57.9%**	**47.8%**
NN	**80.0%**	**65.9%**	**49.6%**	**34.1%**

Result: MSE reductions are larger for more flexible learners (NN and KRR) and generally decrease as n_0 increases.

Monthly Active User Prediction at Meta



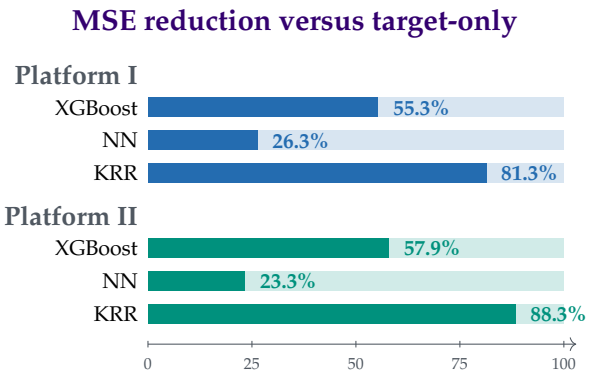
Objective: Predict country-level MAU for a held-out app.

- Leave-one-app-out: four source apps; ≈ 200 observations per app.
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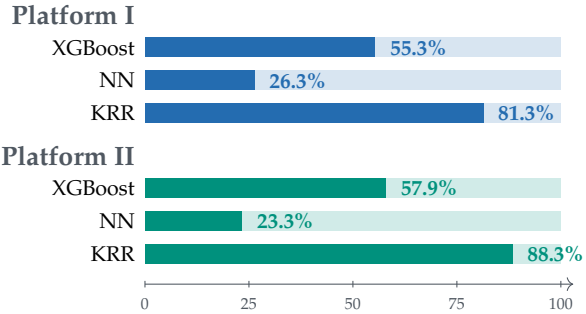
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MSE reduction versus target-only



Average test MSE (30 Monte Carlo runs)

	TLCQM			TKRR	CDAR	DARC
	XGBoost	NN	KRR			
Platform I	**4.099 (0.014)**	**4.211 (0.012)**	**4.073 (0.014)**	6.981	139.327 (1.093)	18.220 (1.156)
Platform II	**3.785 (0.065)**	**3.840 (0.046)**	**3.767 (0.068)**	4.967	117.149 (1.194)	11.542 (0.921)

Standard errors in parentheses, and values below 0.001 are omitted.

Main Takeaways of TLCQM

1. **Generate conditional distributions,** from each source at target covariates.

2. **Calibrate with conditional CRPS** to align the source-generated and target response distributions.

3. **Train a standard predictor** on a target-aligned augmented dataset.

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Thank you!

More details are in

https://arxiv.org/abs/2602.02358.

Bottom Line: TLCQM improves target prediction when conditional transfer bias is small, but persistent bias can cause negative transfer.

References

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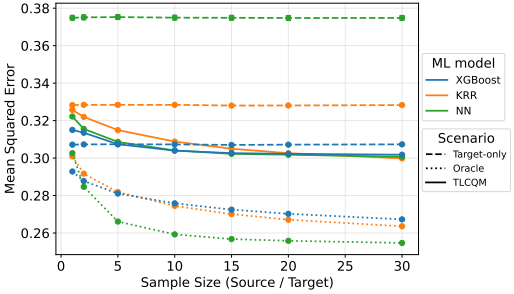
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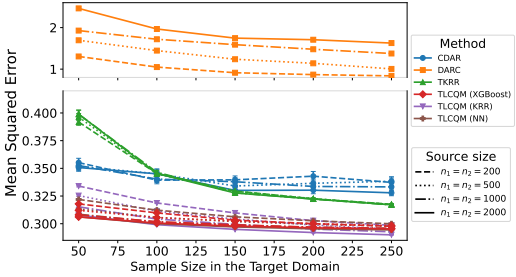
- *Two Source Domains:*

$$Y^{(1)} = \sin\left(3\theta^T X^{(1)}\right) + 1 + \epsilon, \quad Y^{(2)} = 2\cos\left(3\theta^T X^{(2)}\right) + 1 + \epsilon,$$

where $\theta = \left(1, \frac{1}{2}, \ldots, \frac{1}{5}\right)^T \in \mathbb{R}^5$, $X^{(1)}, X^{(2)} \sim \mathcal{N}(\mathbf{1}_5, \boldsymbol{I}_5)$, and $\epsilon \sim \mathcal{N}(0, 0.25)$.

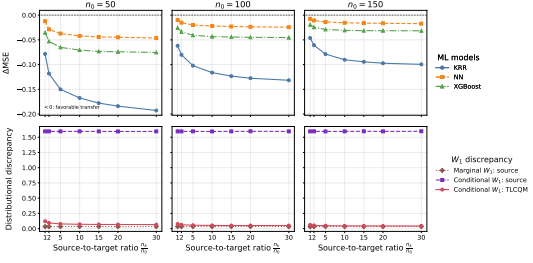
- *Target Domain:* $Y^{(0)} = \frac{1}{3}\sin\left(3\theta^T X^{(0)}\right) - 3 + \epsilon$ with $X^{(0)} \sim \mathcal{N}(\mathbf{0}_5, 0.25 \cdot \boldsymbol{I}_5)$ and $\epsilon \sim \mathcal{N}(0, 0.25)$.



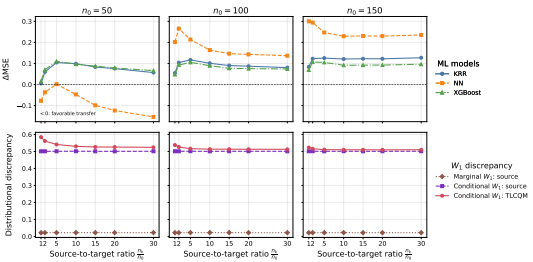


Diagnostics: When Does Transfer Help?

Reversed conditional relation



Target outside the affine span



Conditional alignment, not marginal similarity, determines transfer: TLCQM corrects a reversible conditional mismatch, whereas persistent bias outside the calibrated family can produce negative transfer.